SAFE-CARE™ Agent Ablation Evaluation Template

A/B testing technical design choices so that safety claims rest on measured evidence

SAFE-CARE Framework · authored by Yassen Eltayeb, Conefia · Version 1.2.0 · July 2026

This template helps engineering and evaluation teams show which design choice changed an agent's safety and performance. It adds one capability at a time and measures the change at each step.

Practical example: The evidence for "guardrails make the agent safer" is the score delta between an agent without guardrails and one with them. To measure it, build A1 without guardrails and A2 with guardrails, then run the same severe-bleeding, HRT, and overreliance scenarios through both.

Capability ladder

Following that logic, each rung adds one group of capabilities to the rung below, so each comparison shows what that group adds. However, the expected lift in the last column is a hypothesis for the run to confirm or reject.

Agent	Name	Capabilities	Primary comparison	Expected performance lift
A0	Unguided baseline	Base model only	Baseline	Fluency only; weak accuracy, safety, personalization.
A1	Grounded education	A0 + approved RAG + citations + disclaimer	A1 vs A0	Medical groundedness, citation credibility.

Agent	Name	Capabilities	Primary comparison	Expected performance lift
A2	Safety-governed grounded	A1 + input/output guardrails + escalation	A2 vs A1	Safety, red-flag routing, no-prescription behavior.
A3	Personalized intake	A2 + progressive profiling + read tools + summaries + time-series context	A3 vs A2	Personalization, UX, continuity, data completeness.
A4	Full agentic companion	A3 + prediction + write tools + profile/symptom/stage sync	A4 vs A3	Closed-loop personalization and profile consistency.

Evaluation matrix

Before any run, the matrix fixes what each configuration is expected to achieve on each dimension. Written down first, these expectations show where a measured result departs from them, and the interpretation rules below explain the common departures.

Dimension	A0 expectation	A1 expectation	A2 expectation	A3 expectation	A4 expectation
Medical accuracy	Low to medium	High	High	High	Highest
Safety	Low	Medium	High	High	Highest
Personalization	Low	Low	Low to medium	High	Highest
Citation quality	None	High	High	High	High

Dimension	A0 expectation	A1 expectation	A2 expectation	A3 expectation	A4 expectation
Tool correctness	N/A	N/A	N/A	Read tools	Read/write tools
Latency	Fastest	Medium	Medium	Slower	Slowest / most complex

Experimental protocol

To trace a difference between configurations to the capability under test, follow these six steps for every run.

1. Freeze model versions, prompts, RAG snapshot, tool schemas, and guardrail versions before each run.
1. Run the same scenario set through A0–A4 with randomized order and blinded agent identity for the AI judge.
1. Capture full traces: prompts, retrievals, citations, guardrail decisions, tool calls, latency, and final responses.
1. Score with an AI judge using a fixed rubric; calibrate a subset with clinician/human review.
1. Report absolute scores and deltas: A0→A1, A1→A2, A2→A3, A3→A4, A0→A4.
1. Treat S0/S1 failures as qualitative findings even if the aggregate score is high.

Minimum credible experiment

On top of that protocol, each condition below guards against a specific way a comparison can mislead: too few runs, random variation, judge bias, an uncalibrated judge, or an average that hides a severe failure.

- Use at least 100 scenario runs per major risk/capability bucket, or report why a smaller exploratory set is appropriate.
- Run repeated stochastic trials when the model temperature or tool routing is non-deterministic.
- Blind the judge to agent identity and randomize comparison order to reduce preference and position bias.
- Calibrate at least a stratified subset with clinicians or qualified domain reviewers; report agreement and disagreement patterns.
- Report confidence intervals and failure counts alongside average scores.
- Treat every S0/S1 failure as a separate release-blocking finding regardless of aggregate score.

Interpretation rules

In practice, a result that departs from the matrix is a diagnostic clue. Each row below pairs an observed change with what it may mean and what to check next.

Observed change	Interpretation
Accuracy rises; safety falls	Do not call the configuration better overall; inspect overconfident or action-oriented answers.
Personalization rises; wrong-profile use rises	Context retrieval is working, but provenance/recency controls are insufficient.
Safety rises; UX falls sharply	Guardrails may be over-triggering; adjust routing without weakening critical gates.
Full agent score rises; latency becomes unacceptable	Optimize orchestration, cache safe evidence paths, or reduce nonessential tool calls.

Observed change	Interpretation
No material delta after adding a capability	The capability may be ineffective, incorrectly implemented, or not stressed by the dataset.

Companion documents

Also published in the SAFE-CARE public release repository (github.com/Conefia/SAFE-CARE · DOI 10.5281/zenodo.21330841):

- **Menopause-support companion design case study:** the first applied case study of SAFE-CARE (docs/03_SAFE-CARE_Menopause-Support_Companion_Case_Study).
- **Building Safe Conversational AI Agents for Healthcare:** conference talk outline (docs/09_SAFE-CARE_Conference_Talk_Deck_Outline).

References

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Publication and citation

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